LUCAS WILLIAM JUNGES

Systems Analyst at WEG Energia | Control & Automation Engineer | SAP Engineering BOM Automation

Brazilian citizen · Italian citizenship in final recognition stage (jure sanguinis)

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PROFILE

Systems analyst at **WEG Energia** — the engineered-to-order machines division of WEG, the world's largest low-voltage electric motor manufacturer. I develop and maintain the **SAP-based engineering BOM automation** system for large electric machines: configuration rules, technical lists, and the data models behind them (same problem space as **SAP PLM** / **variant configuration**). Background combines 10+ years of software engineering (Python, SQL, Delphi) with a control & automation degree and hands-on industrial work: PLC integration, machine retrofits, and production ML.

CORE COMPETENCIES

SAP: ECC, S/4HANA, SAP HANA, NetWeaver Java, Solution Manager, BOM/super-BOM, configuration rules, material master | **Databases:** Oracle, SAP HANA, SQL Server, PostgreSQL, MongoDB | **Software:** Python, SQL, C/C++, JavaScript, Delphi, REST APIs, Docker, Git, Linux | **Industrial:** Modbus RTU/TCP, OPC UA, MQTT, PLC integration, SCADA, Raspberry Pi/ESP32 | **ML:** TensorFlow, PyTorch, computer vision, predictive maintenance, edge deployment | **DevOps:** GitHub Actions, Terraform, AWS

PROFESSIONAL EXPERIENCE

Systems Analyst — Engineering Systems *WEG Energia, Jaraguá do Sul, Brazil — May 2026 – Present*

- Development and maintenance of the rule-based **engineering BOM generation system in SAP** for engineered-to-order motors and generators (BOMs of thousands of items per machine)
- Data modeling and relations serving the engineering directorate: configuration rules, validity conditions per machine ID/material ID/rule group, technical specifications
- Backoffice and systems support for engineering; analysis of **SolidWorks** machine and motor-part drawings to model product data

Industrial Automation Engineer *W&Co, Blumenau, Brazil — Nov 2025 – May 2026*

- Retrofit of legacy industrial machines; built a **web HMI** on Raspberry Pi communicating via **Modbus RTU** (RS485) with the machine PLC — still in production

R&D — Industrial IoT & Machine Learning *GreyLogix / 3Fi, Brazil — Apr – Nov 2025*

- Predictive maintenance R&D; **Edge AI** with Google Coral TPU (<100 ms inference), Node-RED/MQTT sensor pipelines, MLOps CI/CD with GitHub Actions, Terraform and AWS

Software Developer *Paytrack / Senior Sistemas, Blumenau, Brazil — 2016 – 2023*

- Corporate travel-expense SaaS: **Python**/**Django**, REST APIs, PostgreSQL, ETL pipelines, AWS infrastructure

Delphi Developer *Operacional Solution, Blumenau, Brazil — 2012 – 2015*

- ERP modules for textile manufacturing (Delphi XE, **Oracle PL/SQL**), in daily factory-floor use

EDUCATION

B.Eng. Control and Automation Engineering *UFSC, Brazil — Completed Dec 2025*

Thesis: Predictive fault detection in three-phase induction motors using infrared thermography & CNNs — **0.912 ROC AUC** (MobileNetV2 + Grad-CAM, deployed on Jetson Xavier with OPC UA/MQTT) — UFSC repository

Technical Degree — Software Development *CEDUP Hermann Hering — 2010 – 2013*

SELECTED PROJECTS

Web HMI, industrial rebar bender: Modbus RTU/RS485, Python asyncio, WebSocket — GitHub | **Edge AI system:** Coral TPU, Docker, MQTT — GitHub | **MLOps pipeline:** GitHub Actions, Terraform, AWS — GitHub

ADDITIONAL INFORMATION

Languages: Portuguese (native), English (C1 — lived in Dublin, 2021), Italian (basic) • **Business entity:** Blumenau TI, CNPJ 53.700.754/0001-10 — B2B contracting, W-8BEN-E available • **Certifications:** Advanced English (Erin School, Dublin), Python Professional Qualification (Entra21), Oracle SQL/PL-SQL • **Portfolio:** lucaswjunges.github.io